Department of Computer Science

Investigating Physics-guided Deep Learning frameworks for subgrid-scale parameterization of Turbulence in Atmospheric Models

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Introduction

Large Eddy Simulation (LES):

- Directly **resolves large turbulent eddies**, responsible for most of the turbulent energy and transport.
- Parameterizes the **small-scale turbulent features using subgrid-scale models (SGS)** (e.g. **Smagorinsky**).

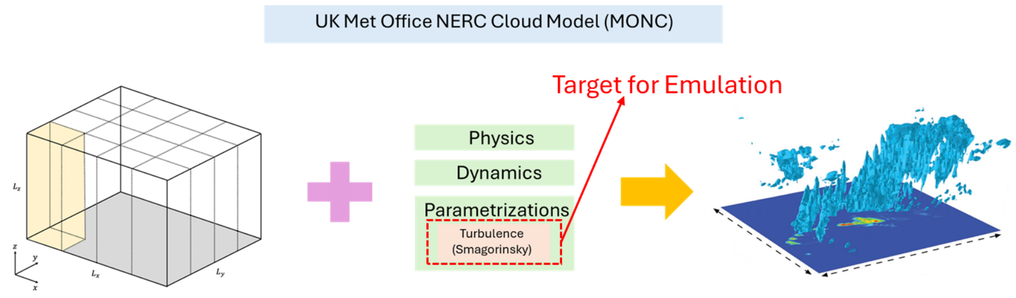


Fig 1: Schematic Representation of the LES setup

Motivation

- Traditional turbulence models: computationally expensive; may not be accurate!
- **Artificial Intelligence (AI)** techniques have shown promise [Beucler et al. (2023), Guan et al. (2023)].
- **Our Approach:** AI based Emulation of SGS Parameterization
- **Objective:** Improve the computational performance

Data and Methodology

Input

3D Data Input
NETCDF files
Channels: Resolved state variables $(zu, zv, zw, zq_{clm}, zq_{wv}, zth)$
[Files x 6 x Z x H x W]

Pre-processing

Vertical Height (VH)-Informed Slicing
2D Samples
[B, 6+VH, H, W]

Key PGDM Features
- ✓ VH Informed 2D processing (memory efficient)
- ✓ Multi-scale feature extraction (3x3, 5x5, 7x7)
- ✓ Physics-guided (Loss + Constraints)
- ✓ Scalable Training with support for DDP and mixed precision
- ✓ Flexible support for normalization and interpolation

Physics-guided 2D CNN based Model (PGDM)

PGDM Architecture
PGDM Core
3x3, 5x5, 7x7 → Residual Blocks + GELU +Dropout → 1x1 Conv → 5 Output channels
2D CNN based: Parallel conv + Residual Connections

Physics Constraint Layer
Ensure Positive Coefficients
Softplus Activation
[B, 5, H, W]

SGS Outputs
Tendencies (viscosity) and Coefficients (Diffusion and Viscosity)
[B, 5, H, W]

Physics-Informed Loss Function
MSE Loss: MSE (tend) + MSE (coeff)
Energy Loss (Penalizes positive energy creation)
Gradient Loss: To preserve gradients/fine structures
Weighted sum of individual loss components

Training Configuration

Batch Size	Epochs	Optimizer	LR with Scheduler	Mixed Precision	Multi-GPU	Weight Decay	Dropout	Normalization	Interpolation
[512, 6+VH, 64, 64]	[300]	[ADAM]	[0.001]	[ON/OFF]	[ON/OFF]	[1e-3]	[0.3]	[ON/OFF]	[ON/OFF]

Fig 2: Summary of the PGDM model, data pre-processing and the training set up.

Training/Validation/Testing Data Set:

3-D diagnostics data from **MONC:** Different idealized simulations (e.g. Radiative Convective Equilibrium (RCE))

- **RCE Simulations**, **sampling frequency (SF) = 1800 sec**
  - **Inputs:** Wind velocity (zu, zv, zw), mass mixing ratios (zq_{clm}, zq_{wv}), potential temperature perturbations (zth)
  - **Targets:** Tendency fields for $(u_{tend}, v_{tend}, w_{tend})$, eddy viscosity $(visc)$ and diffusivity $(diff)$ coefficients

Results

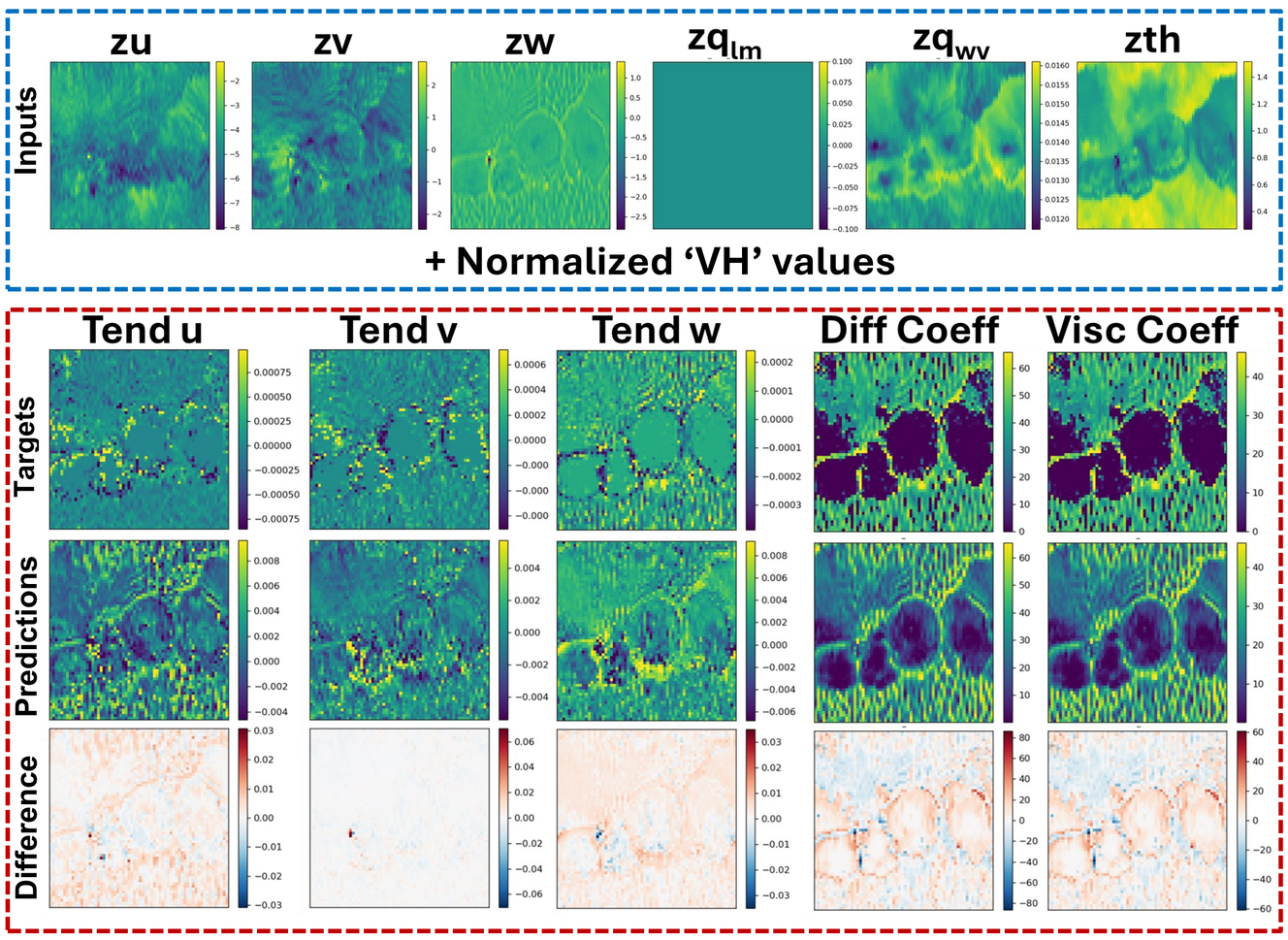
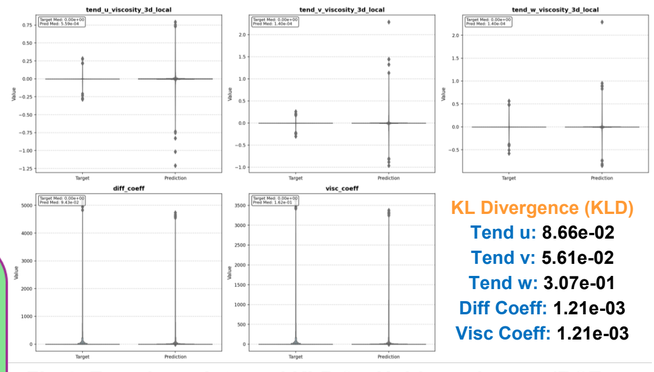


Fig 3: Sample results from validation on hold-out data sets (RCE, SF=1800 sec, random 2D slice)



KL Divergence (KLD)
Tend u: 8.66e-02
Tend v: 5.61e-02
Tend w: 3.07e-01
Diff Coeff: 1.21e-03
Visc Coeff: 1.21e-03

Fig 4: Error box plots and KLD for Hold-out dataset (RCE, SF=1800 sec, random 2D slices from shuffled data)

Variable	MSE	MAE	Correlation	R2
Tend u	5.54e-06	0.001	0.071	-1095.50
Tend v	3.04e-06	0.001	0.001	-125.93
Tend w	4.21e-06	0.001	0.052	-318.94
Diff Coeff	2.71e+02	3.479	0.495	0.13
Visc Coeff	1.34e+02	2.526	0.494	0.12

Table 1: Evaluation Metrics for Unseen Data (RCE, SF=200 sec, from fully reconstructed 3D volume after inference)

Concluding remarks

- **Multi-variable prediction:** Physics-guided AI model (PGDM), with additional VH context demonstrated successful emulation with decent accuracy.
- **Prediction accuracy:** Spatial structures/patterns captured well, poor capture of magnitudes and data distribution. Occasionally produces weird outliers.
- **Prediction variability:** Model Accuracy varied across the different target variables, as per the evaluation metrics used.
- **Resolution Invariance and Generalization:** Not achieved in this proof-of-concept. Would require further exploration and investigation.
- **Physics constraints:** Had a positive impact in model training and achieving decent accuracy, even with limited training data set.

Future Work

- Expand training data (**volume and variety**), explore various other advanced AI frameworks and evaluation metrics for model intercomparison.
- Test the **stability** of AI models by **integrating** them with **MONC** for real-time online testing.

References

1. Beucler, T., et al. (2023). Machine Learning for Clouds and Climate, In Clouds and Their Climatic Impacts (eds S.C. Sullivan and C. Hoose), chapter 16.
2. Guan, Y., et al. (2023). Learning physics-constrained subgrid-scale closures in the small-data regime for stable and accurate LES, Physica D: Nonlinear Phenomena, 443, 133568.

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